

# BST Physics Curriculum Vol 12 Ch 6 — Spectroscopy (UV-Vis + IR + NMR + X-ray) v0.4 (Calibration #23 substance-floor refill)

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redirect)

## Vol 12 Chapter 6 — Spectroscopy (UV-Vis + IR + NMR + X-ray) + Substrate Cartography

### Chapter motivation

Spectroscopy is chemistry's primary observational tool: probe matter with electromagnetic radiation across the spectrum (X-ray through radio) and read substrate structure from the absorption/emission/scattering response. Each technique probes a different energy scale: X-ray ( $10^2$ - $10^4$  eV) → electron density (crystallography); UV-Vis (1-10 eV) → electronic transitions; IR (0.01-1 eV) → vibrational transitions; Raman (vibrational, with shifted frequency) → vibrational + rotational; NMR ( $10^{-7}$ - $10^{-6}$  eV) → nuclear spin in magnetic field. Standard chemistry treatment: Beer-Lambert law  $A = \epsilon c l$ ; selection rules from quantum mechanics ( $\Delta l = \pm 1$  electric dipole;  $\Delta v = \pm 1$  vibrational harmonic;  $\Delta m_l = \pm 1$  NMR); transition strengths from matrix elements. BST substrate-cartography reading: spectroscopy IS substrate-readout — selection rules are substrate K-type representation transitions; transition strengths follow  $\alpha^{\{\text{BST primary}\}}$  pattern per T2476 (substrate fine-structure constant emerges in transition matrix elements); X-ray diffraction reads substrate periodic structure tied to BST primary integer ladder.

### Section 6.0b — Reader-grade 3-level pedagogy

Level 1: Spectroscopy = substrate-readout. UV-Vis probes electronic K-types; IR probes vibrational K-types; NMR probes nuclear-spin K-types; X-ray probes substrate periodic lattice. Selection rules are K-type representation transitions; transition strengths follow  $\alpha^{\{\text{BST primary}\}}$  per T2476.

Level 2 (graduate-chemist accessible): Electromagnetic spectrum spans 18 orders

of magnitude (radio  $10^{-9}$  eV through gamma  $10^9$  eV). Each spectroscopic technique probes a specific energy range. UV-Visible spectroscopy (1-10 eV, 200-800 nm): electronic transitions between MOs; Beer-Lambert law  $A = \epsilon cl$  gives quantitative concentration. Selection rules:  $\Delta l = \pm 1$  (electric dipole), spin-allowed ( $\Delta S = 0$ ), parity (Laporte rule:  $g \leftrightarrow u$  for centrosymmetric molecules). Infrared spectroscopy (0.01-1 eV, 2.5-25  $\mu\text{m}$ ): vibrational transitions; selection rule  $\Delta v = \pm 1$  (harmonic),  $\Delta v = \pm 2, \pm 3$  (overtone); characteristic stretching frequencies (C-H 2900  $\text{cm}^{-1}$ , C=O 1700  $\text{cm}^{-1}$ , O-H 3400  $\text{cm}^{-1}$ ) provide functional group fingerprints. Raman spectroscopy: inelastic photon scattering; Stokes ( $\Delta v = +1$ , photon loses energy) and anti-Stokes ( $\Delta v = -1$ , photon gains energy) shifts; complementary to IR (different selection rules: Raman requires polarizability change, IR requires dipole change). Nuclear Magnetic Resonance (NMR,  $10^{-7}$  eV, radiofrequency): nuclear spin transitions in applied magnetic field; chemical shift  $\delta$  (ppm) probes local electronic environment; J-coupling (Hz) probes through-bond connectivity;  $^1\text{H}$  and  $^{13}\text{C}$  NMR ubiquitous in organic chemistry. X-ray crystallography ( $10^2$ - $10^4$  eV): elastic scattering from electron density; Bragg's law  $n\lambda = 2d \sin \theta$  relates diffraction angles to lattice spacings; full 3D structure from intensities of diffraction spots. BST substrate-cartography reading: each spectroscopic technique reads a different substrate K-type spectrum. Electronic K-types are Wallach K-type representations on  $D_{IV}^5$  with Casimir eigenvalues at electronic energy scale; vibrational K-types are substrate sub-structure modes at vibrational scale; nuclear-spin K-types are substrate  $\frac{1}{2}$ -integer representations at nuclear scale; X-ray probes substrate periodic lattice directly. Selection rules in each case are substrate K-type representation theory:  $\Delta l = \pm 1$  electric dipole selection corresponds to specific substrate K-type tensor product allowed transitions. Per T2476  $\alpha^{\{\text{BST primary}\}}$  substrate-mechanism: transition matrix elements scale with  $\alpha = 1/137.036 = 1/N_{\text{max}}$  from substrate Wallach K-type evaluation; transition strengths follow  $\alpha^{\{\text{BST primary integer}\}}$  pattern depending on multipole order (electric dipole  $\alpha$ ; magnetic dipole  $\alpha^3$ ; electric quadrupole  $\alpha^3$ ).

Level 3 (5th-grader accessible): Spectroscopy is like shining light on stuff and listening to what comes back. Different colors of light find different things: X-rays see the arrangement of atoms in crystals; UV-visible light sees electron jumps between energy levels; infrared light sees molecules vibrating like springs; radio waves (NMR) see atomic nuclei wobbling in magnetic fields. Each tool tells you something different about chemical structure. BST framework says all of this is reading the substrate's K-type structure — different tools read different parts of the substrate “vocabulary.”

## Section 6.1 — UV-Vis Spectroscopy (Electronic Transitions)

Beer-Lambert law:  $A = \epsilon cl$  (absorbance = molar absorptivity  $\times$  concentration  $\times$  path length).

Electronic transitions:  $\sigma \rightarrow \sigma^*$  (vacuum UV  $\sim 200$  nm),  $n \rightarrow \pi^*$  (300-400 nm common),  $\pi \rightarrow \pi^*$  (200-300 nm common),  $d \rightarrow d$  (transition metal colors, visible).

Selection rules:  $\Delta l = \pm 1$  (electric dipole),  $\Delta S = 0$  (spin-allowed), Laporte  $g \leftrightarrow u$  (parity).

BST framing: electronic K-type transitions between Wallach K-type representations on  $D_{IV}^5$  at electronic energy scale.

## Section 6.2 — IR Spectroscopy (Vibrational Transitions)

Selection rule  $\Delta v = \pm 1$  (harmonic oscillator); overtones  $\Delta v = \pm 2, \pm 3$  (anharmonic).

Characteristic frequencies provide functional group identification: - C-H stretch:  $2850\text{--}3300\text{ cm}^{-1}$  - C=O stretch:  $1650\text{--}1850\text{ cm}^{-1}$  - O-H stretch:  $3200\text{--}3650\text{ cm}^{-1}$  (broad if H-bonded) - N-H stretch:  $3300\text{--}3500\text{ cm}^{-1}$

Fingerprint region  $600\text{--}1500\text{ cm}^{-1}$  used for compound identification.

BST framing: vibrational K-types as substrate sub-structure modes; vibrational frequencies = substrate K-type Casimir-eigenvalue differences at vibrational scale.

## Section 6.3 — Raman Spectroscopy

Inelastic photon scattering: Stokes shift ( $\Delta v = +1$ , photon loses energy) and anti-Stokes shift ( $\Delta v = -1$ , photon gains energy).

Complementary to IR: Raman requires polarizability change (centrosymmetric vibrations); IR requires dipole change (asymmetric vibrations). Centrosymmetric molecules (e.g.,  $\text{CO}_2$  symmetric stretch) show in Raman but not IR.

BST framing: Raman probes substrate polarizability-active K-types; IR probes substrate dipole-active K-types; together complete vibrational K-type spectroscopy.

## Section 6.4 — NMR Spectroscopy (Nuclear Spin)

Nuclear spin in applied magnetic field  $B_0$  splits into  $2I+1$  levels (Zeeman effect). For  $I = \frac{1}{2}$  ( $^1\text{H}$ ,  $^{13}\text{C}$ ,  $^{31}\text{P}$ ), two levels separated by  $\Delta E = \gamma \hbar B_0$  where  $\gamma$  is gyromagnetic ratio.

Chemical shift  $\delta$  (ppm) probes local electron environment via shielding constant  $\sigma$ . J-coupling (Hz) probes through-bond connectivity (typically 1-15 Hz for  $^1\text{H}\text{--}^1\text{H}$ ).

BST framing: nuclear spin K-types as substrate  $\frac{1}{2}$ -integer representations; chemical shifts read substrate sub-structure electronic environment.

## Section 6.5 — X-Ray Crystallography

Bragg's law:  $n\lambda = 2d \sin \theta$  relates X-ray wavelength + diffraction angle to lattice spacing  $d$ .

Full 3D structure determination from diffraction intensities  $I(hkl)$  via Fourier transform to electron density  $\rho(r)$ . Phase problem resolved by isomorphous replacement, anomalous dispersion, or direct methods (Hauptman-Karle Nobel 1985).

BST framing: X-ray diffraction reads substrate periodic lattice directly; lattice spacings emerge from BST primary integer structure (Vol 11 Generative Geometry + Topology cross-link).

## Section 6.6 — $\alpha^{\{\text{BST primary}\}}$ Transition Strength Pattern (T2476)

Per T2476 substrate-mechanism Friday 2026-05-22: transition matrix elements scale with  $\alpha = 1/137.036 = 1/N_{\text{max}}$ . Multipole hierarchy: - Electric dipole (E1):  $\sim\alpha^1$  (allowed, strongest) - Magnetic dipole (M1):  $\sim\alpha^3$  (weak) - Electric quadrupole (E2):  $\sim\alpha^3$  (weak) - Higher multipoles:  $\sim\alpha^{(2L-1)}$  for  $E_L$

Substrate origin:  $\alpha^{\{\text{BST primary integer}\}}$  exponent pattern from Wallach K-type evaluation per T2476.

## Section 6.7 — Connection

- Vol 5 Ch 6 atomic spectroscopy (substrate-level treatment)
- Vol 1 Ch 5 Casimir invariants (transition matrix element foundation)
- Vol 11 Generative Geometry + Topology (X-ray lattice structure)
- T2476  $\alpha^{\{\text{BST primary}\}}$  exponent pattern (transition strength substrate-mechanism)
- Vol 12 Ch 2-3 Chemical bonding + MO theory (transition origin)
- Vol 13 Ch 1+7 Biology substrate-reading (NMR + X-ray for biomolecules)

Honest scope: K-type representation framework + T2476  $\alpha^{\{\text{BST primary}\}}$  substrate-mechanism provide organizing principles; specific quantitative spectroscopic intensity calculations from substrate Hamiltonian remain multi-year SP-31 program. T2476 Cal #100 multi-CI ratification in flight.

— Lyra, Vol 12 Ch 6 v0.4 chapter-grade narrative REFILLED per Cal #104 + Calibration #23, Saturday 2026-05-23 EDT